

```
    set type hard-switch
    set stp enable
    set device-identification enable
    set lldp-transmission enable
    set role lan
    set snmp-index 55
    set ip-managed-by-fortiipam disable
  next
end
```

3. Disable STP on port5 by enabling it as an STP edge port:

```
config system interface
  edit "port5"
    set vdom "vdom1"
    set type physical
    set stp-edge enable
    set snmp-index 9
  next
end
```

Port5 is enabled as an edge port with STP disabled. Port3 remains enabled for STP.

Zone

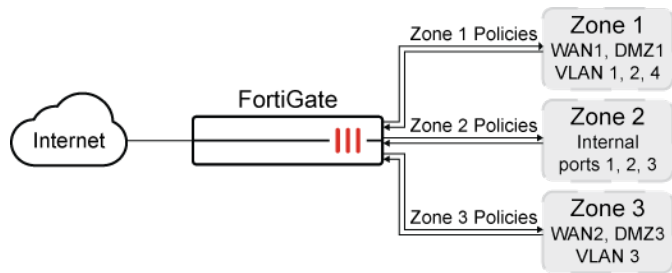
Zones are a group of one or more physical or virtual FortiGate interfaces that you can apply firewall policies to for controlling inbound and outbound traffic. Grouping interfaces and VLAN subinterfaces into zones simplifies creating firewall policies where a number of network segments can use the same policy settings and protection profiles.

When you add a zone, you select the names of the interfaces and VLAN subinterfaces to add to the zone. Each interface still has its own address. Routing is still done between interfaces, that is, routing is not affected by zones. You can use firewall policies to control the flow of intra-zone traffic.

For example, in the sample configuration below, the network includes three separate groups of users representing different entities on the company network. While each group has its own set of ports and VLANs in each area, they can all use the same firewall policy and protection profiles to access the Internet. Rather than the administrator making nine separate firewall policies, he can make administration simpler by adding the required interfaces to a zone and creating three policies.

Example configuration

You can configure policies for connections to and from a zone but not between interfaces in a zone. For this example, you can create a firewall policy to go between zone 1 and zone 3, but not between WAN2 and WAN1, or WAN1 and DMZ1.



To create a zone in the GUI:

1. Go to *Network > Interfaces*.



If VDOMs are enabled, go to the VDOM to create a zone.

2. Click *Create New > Zone*.
3. Configure the *Name* and add the *Interface Members*.
4. Enable or disable *Block intra-zone traffic* as required.
5. Click *OK*.

To configure a zone to include the interfaces WAN1, DMZ1, VLAN1, VLAN2 and VLAN4 using the CLI:

```

config system zone
  edit zone_1
    set interface WAN1 DMZ1 VLAN1 VLAN2 VLAN4
    set intrazone {deny | allow}
  next
end

```

Using zone in a firewall policy

To configure a firewall policy to allow any interface to access the Internet using the CLI:

```

config firewall policy
  edit 2
    set name "2"
    set srcintf "zone_1"
    set dstintf "port15"
    set srcaddr "all"
    set dstaddr "all"
    set action accept
    set schedule "always"
    set service "ALL"
    set nat enable
  next
end

```

Intra-zone traffic

In the zone configuration you can set `intrazone deny` to prohibit the different interfaces in the same zone to talk to each other.

For example, if you have ten interfaces in your zone and the `intrazone` setting is `deny`. You now want to allow traffic between a very small number of networks on different interfaces that are part of the zone but you do not want to disable the intra-zone blocking.

In this example, the zone VLANs are defined as: 192.168.1.0/24, 192.168.2.0/24, ... 192.168.10.0/24.

This policy allows traffic from 192.168.1.x to 192.168.2.x even though they are in the same zone and intra-zone blocking is enabled. The intra-zone blocking acts as a default deny rule and you have to specifically override it by creating a policy within the zone.

To enable intra-zone traffic, create the following policy:

Source Interface	Zone-name, e.g., V1ans
Source Address	192.168.1.0/24
Destination	Zone-name (same as Source Interface, i.e., V1ans)
Destination Address	192.168.2.0/24

Virtual wire pair

A virtual wire pair consists of two interfaces that do not have IP addressing and are treated like a transparent mode VDOM. All traffic received by one interface in the virtual wire pair can only be forwarded to the other interface, provided a virtual wire pair firewall policy allows this traffic. Traffic from other interfaces cannot be routed to the interfaces in a virtual wire pair. Redundant and 802.3ad aggregate (LACP) interfaces can be included in a virtual wire pair.

Virtual wire pairs are useful for a typical topology where MAC addresses do not behave normally. For example, port pairing can be used in a Direct Server Return (DSR) topology where the response MAC address pair may not match the request's MAC address pair.



When creating a new virtual wire pair, the *Interface members* field displays interfaces without assigned addresses. Interfaces with assigned addresses are not displayed.

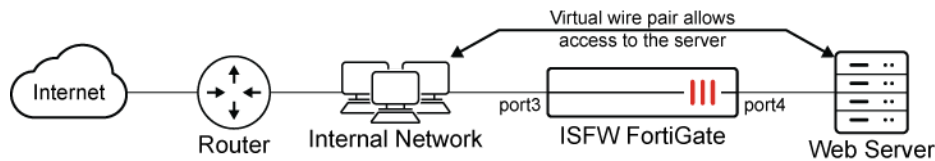
Therefore, you cannot add to a virtual wire pair an interface with *Addressing mode* set to *DHCP*. If you change the interface settings to *Manual* with *IP/Netmask* set to *0.0.0.0/0.0.0.0*, you can add the interface to a virtual wire pair.

Example

In this example, a virtual wire pair (port3 and port4) makes it easier to protect a web server that is behind a FortiGate operating as an Internal Segmentation Firewall (ISFW). Users on the internal network access the web server through the ISFW over the virtual wire pair.



Interfaces used in a virtual wire pair cannot be used to access the ISFW FortiGate. Before creating a virtual wire pair, make sure you have a different port configured to allow admin access using your preferred protocol.



To add a virtual wire pair using the GUI:

1. Go to *Network > Interfaces*.
2. Click *Create New > Virtual Wire Pair*.
3. Enter a name for the virtual wire pair.
4. Select the *Interface Members* to add to the virtual wire pair (*port3* and *port 4*).
These interfaces cannot be part of a switch, such as the default LAN/internal interface.
5. If required, enable *Wildcard VLAN* and set the *VLAN Filter*.
6. Click *OK*.

To add a virtual wire pair using the CLI:

```

config system virtual-wire-pair
  edit "VWP-name"
    set member "port3" "port4"
    set wildcard-vlan disable
  next
end
  
```

To create a virtual wire pair policy using the GUI:

1. Go to *Policy & Objects > Firewall Virtual Wire Pair Policy*.
2. Click *Create New*.
3. In the *Virtual Wire Pair* field, click the *+* to add the virtual wire pair.
4. Select the direction (arrows) that traffic is allowed to flow.
5. Configure the other settings as needed.
6. Click *OK*.

To create a virtual wire pair policy using the CLI:

```

config firewall policy
  edit 1
    set name "VWP-Policy"
    set srcintf "port3" "port4"
    set dstintf "port3" "port4"
    set srcaddr "all"
    set dstaddr "all"
  
```

```

        set action accept
        set schedule "always"
        set service "ALL"
        set utm-status enable
        set fsso disable
    next
end

```

Configuring multiple virtual wire pairs in a virtual wire pair policy

You can create a virtual wire pair policy that includes different virtual wire pairs in NGFW profile and policy mode. This reduces overhead to create multiple similar policies for each VWP. In NGFW policy mode, multiple virtual wire pairs can be configured in a *Security Virtual Wire Pair Policy* and *Virtual Wire Pair SSL Inspection & Authentication* policy.

The virtual wire pair settings must have wildcard VLAN enabled. When configuring a policy in the CLI, the virtual wire pair members must be entered in `srcintf` and `dstintf` as pairs.

To configure multiple virtual wire pairs in a policy in the GUI:

1. Configure the virtual wire pairs:

- a. Go to *Network > Interfaces* and click *Create New > Virtual Wire Pair*.
- b. Create a pair with the following settings:

Name	test-vwp-1
Interface members	wan1, wan2
Wildcard VLAN	Enable

- c. Click *OK*.
- d. Click *Create New > Virtual Wire Pair* and create another pair with the following settings:

Name	test-vwp-2
Interface members	port19, port20
Wildcard VLAN	Enable

- e. Click *OK*.

2. Configure the policy:

- a. Go to *Policy & Objects > Firewall Virtual Wire Pair Policy* and click *Create New*.
- b. In the *Virtual Wire Pair* field, click the **+** to add *test-vwp-1* and *test-vwp-2*. Select the direction for each of the selected virtual wire pairs.

Create New Policy

ID: 1

Name:

Virtual Wire Pair: test-vwp-1, test-vwp-2

Source: wan1, port19

Destination: wan2 (test-vwp-1), port20 (test-vwp-2)

Schedule: always

Service:

Action: ☒ ACCEPT ☐ DENY

Inspection Mode: ☒ Flow-based ☐ Proxy-based

Firewall/Network Options

NAT: ☐

Protocol Options: PROT default

OK Cancel

- c. Configure the other settings as needed.
- d. Click OK.

To configure multiple virtual wire pairs in a policy in the CLI:

1. Configure the virtual wire pairs:

```
config system virtual-wire-pair
  edit "test-vwp-1"
    set member "wan1" "wan2"
    set wildcard-vlan enable
  next
  edit "test-vwp-2"
    set member "port19" "port20"
    set wildcard-vlan enable
  next
end
```

2. Configure the policy:

```
config firewall policy
  edit 1
    set name "vwp1&2-policy"
    set srcintf "port19" "wan1"
    set dstintf "port20" "wan2"
    set srcaddr "all"
    set dstaddr "all"
    set action accept
    set schedule "always"
    set service "ALL"
    set logtraffic all
  next
end
```

PRP handling in NAT mode with virtual wire pair

PRP (Parallel Redundancy Protocol) is supported in NAT mode for a virtual wire pair. This preserves the PRP RCT (redundancy control trailer) while the packet is processed by the FortiGate.

To configure PRP handling on a device in NAT mode:

1. Enable PRP in the VDOM settings:

```
(root) # config system settings
      set prp-trailer-action enable
end
```

2. Enable PRP in the NPU attributes:

```
(global) # config system npu
      set prp-port-in "port15"
      set prp-port-out "port16"
end
```

3. Configure the virtual wire pair:

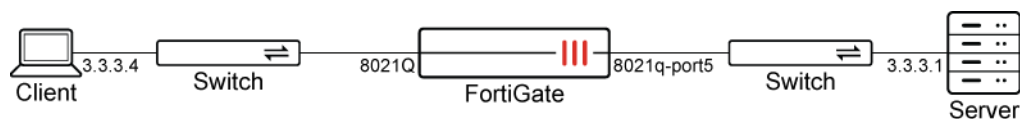
```
(root) # config system virtual-wire-pair
      edit "test-vwp-1"
        set member "port15" "port16"
      next
end
```

Using VLAN sub-interfaces in virtual wire pairs

VLAN sub-interfaces, such as regular 802.1Q and 802.1ad (QinQ), are allowed to be members of a virtual wire pair.

Example

In this example, the FortiGate has two VLAN interfaces. The first interface is a QinQ (802.1ad) interface over the physical interface port3. The second interface is a basic 802.1Q VLAN interface over physical interface port5. These two interfaces are grouped in a virtual wire pair so that bi-directional traffic is allowed. This example demonstrates ICMP from the client (3.3.3.4) sent to the server (3.3.3.1).



To configure VLAN sub-interfaces in a virtual wire pair:**1. Configure the QinQ interfaces:**

```
config system interface
  edit "8021ad-port3"
    set vdom "vdom1"
    set vlan-protocol 8021ad
    set device-identification enable
    set role lan
    set snmp-index 31
    set interface "port3"
    set vlanid 3
  next
  edit "8021Q"
    set vdom "vdom1"
    set device-identification enable
    set role lan
    set snmp-index 32
    set interface "8021ad-port3"
    set vlanid 33
  next
end
```

2. Configure the 802.1Q interface:

```
config system interface
  edit "8021q-port5"
    set vdom "vdom1"
    set device-identification enable
    set role lan
    set snmp-index 33
    set interface "port5"
    set vlanid 5
  next
end
```

3. Configure the virtual wire pair:

```
config system virtual-wire-pair
  edit "VWP1"
    set member "8021Q" "8021q-port5"
  next
end
```

4. Configure the firewall policy:

```
config firewall policy
  edit 1
    set name "1"
    set srcintf "8021Q" "8021q-port5"
    set dstintf "8021Q" "8021q-port5"
    set action accept
```

```

        set srcaddr "all"
        set dstaddr "all"
        set schedule "always"
        set service "ALL"
    next
end

```

To verify that bi-directional traffic passes through the FortiGate:

```

# diagnose sys session filter policy 1
# diagnose sys session list

session info: proto=1 proto_state=00 duration=18 expire=42 timeout=0 flags=00000000 socktype=0
sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
class_id=0 ha_id=0 policy_dir=0 tunnel=/ vlan_cos=0/0
state=may_dirty br npu
statistic(bytes/packets/allow_err): org=168/2/1 reply=168/2/1 tuples=2
tx speed(Bps/kbps): 0/0 rx speed(Bps/kbps): 0/0
orgin->sink: org pre->post, reply pre->post dev=56->55/55->56 gwy=0.0.0.0/0.0.0.0
hook=pre dir=org act=noop 3.3.3.4:3072->3.3.3.1:8(0.0.0.0:0)
hook=post dir=reply act=noop 3.3.3.1:3072->3.3.3.4:0(0.0.0.0:0)
src_mac=08:5b:0e:71:bf:c6 dst_mac=d4:76:a0:5d:b2:de
misc=0 policy_id=1 pol_uuid_idx=534 auth_info=0 chk_client_info=0 vd=3
serial=00005f6c tos=ff/ff app_list=0 app=0 url_cat=0
rpdb_link_id=00000000 ngfwid=n/a
npu_state=0x4000c00 ofld=0 ofld-R
npu info: flag=0x81/0x81, offload=8/8, ips_offload=0/0, epid=187/156, ipid=156/187,
vlan=0x0005/0x0021
vlifid=156/187, vtag_in=0x0005/0x0021 in_npu=1/1, out_npu=1/1, fwd_en=0/0, qid=0/5
total session 1

```

DVLAN QinQ on NP7 platforms over virtual wire pairs

DVLAN 802.1ad and 802.1Q modes are supported on NP7 platforms over virtual wire pairs, which provides better performance and packet processing.

The default DVLAN mode is 802.1ad, but the DVLAN mode can be changed using `diagnose npu np7 dvlan-mode <dvlan_mode> {<npid> | all}`. The DVLAN mode can be applied to a specific NPID or all NPIDs. For example:

- `diagnose npu np7 dvlan-mode 802.1AD 0` will set NP0 to work in 802.1ad mode.
- `diagnose npu np7 dvlan-mode 802.1Q all` will set all NPUs to work in 802.1Q mode.



A reboot is required for custom DVLAN settings to take effect. To avoid any inconveniences or disruptions, changing the DVLAN settings should be done during a scheduled downtime or maintenance window.

The DVLAN mode should only be changed if you are solely using the virtual wire pair (VWP) and are seeking to enhance performance. Enabling this feature may impact VLAN interfaces within your network.

In the virtual wire pair settings, the `outer-vlan-id` can be set. This is the same value as the outer provider-tag (S-Tag).

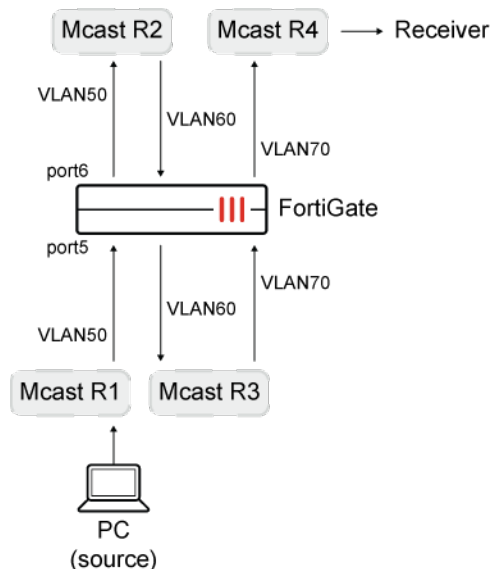
To configure the outer VLAN ID:

```
config system virtual-wire-pair
  edit "dvlan-test"
    set member "port33" "port34"
    set wildcard-vlan enable
    set outer-vlan-id 1234
  next
end
```

Enhanced multicast session key for virtual wire pair **NEW**

FortiOS now adds ingress interface and VLAN ID as additional session keys for multicast traffic in virtual wire pair (VWP) configurations with `wildcard-vlan` enabled. This enhancement enables the FortiGate to create distinct multicast sessions for traffic returning through the same VWP on different VLANs, improving accuracy and ensuring optimal session handling.

Example



In this example, a VWP with wildcard-vlan enabled is created. A firewall policy and multicast policy are created. Multicast traffic goes through FortiGate multiple times in different directions, and distinct multicast sessions are created for traffic returning through the same VWP on different VLANs.

To configure:

1. Configure a virtual wire pair:

The VWP contains members a and b, and *wildcard-vlan* is enabled.

```
config system virtual-wire-pair
  edit "vwp"
    set member "a" "b"
    set wildcard-vlan enable
  next
end
```

2. Configure a firewall policy:

The firewall policy specifies VWP members a and b as source and destination interfaces.

```
config firewall policy
  edit 1
    set uuid dfb4861a-f6ff-51f0-888b-682470ce488d
    set srcintf "a" "b"
    set dstintf "b" "a"
    set action accept
    set srcaddr "all"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
  next
end
```

3. Configure a multicast policy:

```
config firewall multicast-policy
  edit 1
    set uuid 9d6c29a4-f720-51f0-f60b-07a30ec0aaac
    set srcintf "any"
    set dstintf "any"
    set srcaddr "all"
    set dstaddr "all"
    set logtraffic all
  next
end
```

4. After multicast traffic flows, view the different multicast sessions in traffic logs:

```
1: date=2026-01-21 time=15:38:42 eventtime=1769038717948452847 tz="-0800" logid="0002000012"
type="traffic" subtype="multicast" level="notice" vd="root" srcip=10.1.1.1 identifier=13
srcintf="b" srcintfrole="undefined" dstip=225.1.1.1 dstintf="a" dstintfrole="undefined"
vlan=60 srccountry="Reserved" dstcountry="Reserved" sessionid=4089 proto=1 action="accept"
policyid=1 policytype="multicast-policy" poluid="9d6c29a4-f720-51f0-f60b-07a30ec0aaac"
```

```

service="PING" trandisp="noop" appcat="unscanned" duration=181 sentbyte=168 rcvdbyte=0
sentpkt=2 rcvdpkt=0

2: date=2026-01-21 time=15:38:42 eventtime=1769038717948449960 tz="-0800" logid="0002000012"
type="traffic" subtype="multicast" level="notice" vd="root" srcip=10.1.1.1 identifier=13
srcintf="a" srcintfrole="undefined" dstip=225.1.1.1 dstintf="b" dstintfrole="undefined"
vlan=50 srccountry="Reserved" dstcountry="Reserved" sessionid=4088 proto=1 action="accept"
policyid=1 policytype="multicast-policy" poluuid="9d6c29a4-f720-51f0-f60b-07a30ec0aaac"
service="PING" trandisp="noop" appcat="unscanned" duration=181 sentbyte=168 rcvdbyte=0
sentpkt=2 rcvdpkt=0

3: date=2026-01-21 time=15:38:42 eventtime=1769038717940450742 tz="-0800" logid="0002000012"
type="traffic" subtype="multicast" level="notice" vd="root" srcip=10.1.1.1 identifier=13
srcintf="a" srcintfrole="undefined" dstip=225.1.1.1 dstintf="b" dstintfrole="undefined"
vlan=70 srccountry="Reserved" dstcountry="Reserved" sessionid=4090 proto=1 action="accept"
policyid=1 policytype="multicast-policy" poluuid="9d6c29a4-f720-51f0-f60b-07a30ec0aaac"
service="PING" trandisp="noop" appcat="unscanned" duration=181 sentbyte=168 rcvdbyte=0
sentpkt=2 rcvdpkt=0

```

5. View the different multicast sessions using the diagnose sys mcast-session list command:

```

# diagnose sys mcast-session list

session info: id=4264 vf=0 vrf=0 proto=1 10.1.1.1.16->225.1.1.1.8
used=2 path=1 duration=167 expire=13 indev=11 vlanid=50 pkts=2 bytes=168
state=0000001a: tp npu-cap offloaded
session-npu-info: ipid/vlifid=9/9 vlanid/vtag_in=50/50 in_npuuid=1 tae_index=0 qid=1 fwd_
map=0x00000000
path: log offloaded policy=1, outdev=12, tos=0xff
act-npu-info: ipid/vlifid=9/9 vlanid/vtag_in=50/50 in_npu_id=1, out_npuuid=1 epid=10 fwd=0

session info: id=4265 vf=0 vrf=0 proto=1 10.1.1.1.16->225.1.1.1.8
used=2 path=1 duration=167 expire=13 indev=12 vlanid=60 pkts=2 bytes=168
state=0000001a: tp npu-cap offloaded
session-npu-info: ipid/vlifid=10/10 vlanid/vtag_in=60/60 in_npuuid=1 tae_index=0 qid=2 fwd_
map=0x00000000
path: log offloaded policy=1, outdev=11, tos=0xff
act-npu-info: ipid/vlifid=10/10 vlanid/vtag_in=60/60 in_npu_id=1, out_npuuid=1 epid=9 fwd=0

session info: id=4266 vf=0 vrf=0 proto=1 10.1.1.1.16->225.1.1.1.8
used=2 path=1 duration=167 expire=13 indev=11 vlanid=70 pkts=2 bytes=168
state=0000001a: tp npu-cap offloaded
session-npu-info: ipid/vlifid=9/9 vlanid/vtag_in=70/70 in_npuuid=1 tae_index=0 qid=7 fwd_
map=0x00000000
path: log offloaded policy=1, outdev=12, tos=0xff
act-npu-info: ipid/vlifid=9/9 vlanid/vtag_in=70/70 in_npu_id=1, out_npuuid=1 epid=10 fwd=0
Total 3 sessions

```

Enhanced MAC VLAN

The Media Access Control (MAC) Virtual Local Area Network (VLAN) feature in Linux allows you to configure multiple virtual interfaces with different MAC addresses (and therefore different IP addresses) on a physical interface.

FortiGate implements an enhanced MAC VLAN consisting of a MAC VLAN with bridge functionality. Because each MAC VLAN has a unique MAC address, virtual IP addresses (VIPs) and IP pools are supported, and you can disable Source Network Address Translation (SNAT) in policies.

MAC VLAN cannot be used in a transparent mode virtual domain (VDM). In a transparent mode VDM, a packet leaves an interface with the MAC address of the original source instead of the interface's MAC address. FortiGate implements an enhanced version of MAC VLAN where it adds a MAC table in the MAC VLAN which learns the MAC addresses when traffic passes through.

If you configure a VLAN ID for an enhanced MAC VLAN, it won't join the switch of the underlying interface. When a packet is sent to this interface, a VLAN tag is inserted in the packet and the packet is sent to the driver of the underlying interface. When the underlying interface receives a packet, if the VLAN ID doesn't match, it won't deliver the packet to this enhanced MAC VLAN interface.



When using a VLAN ID, the ID and the underlying interface must be a unique pair, even if they belong to different VDMs. This is because the underlying, physical interface uses the VLAN ID as the identifier to dispatch traffic among the VLAN and enhanced MAC VLAN interfaces.

If you use an interface in an enhanced MAC VLAN, do not use it for other purposes such as a management interface, HA heartbeat interface, or in Transparent VDMs.

If a physical interface is used by an EMAC VLAN interface, you cannot use it in a Virtual Wire Pair.

In high availability (HA) configurations, enhanced MAC VLAN is treated as a physical interface. It's assigned a unique physical interface ID and the MAC table is synchronized with the secondary devices in the same HA cluster.

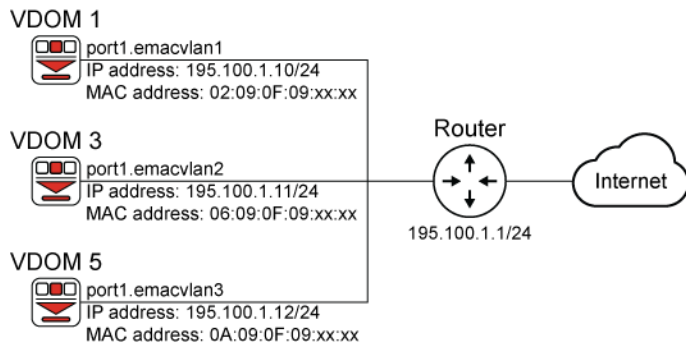


In HA configurations, FortiGate assigns a virtual MAC to each interface. Virtual interfaces, such as EMAC VLAN interfaces with underlying NPU VLINK interface, are an exception and do not get assigned virtual MAC addresses.

Example 1: Enhanced MAC VLAN configuration for multiple VDMs that use the same interface or VLAN

In this example, a FortiGate is connected, through port 1 to a router that's connected to the Internet. Three VDMs share the same interface (port 1) which connects to the same router that's connected to the Internet. Three enhanced MAC VLAN interfaces are configured on port 1 for the three VDMs. The enhanced MAC VLAN interfaces are in the same IP subnet segment and each have unique MAC addresses.

The underlying interface (port 1) can be a physical interface, an aggregate interface, or a VLAN interface on a physical or aggregate interface.



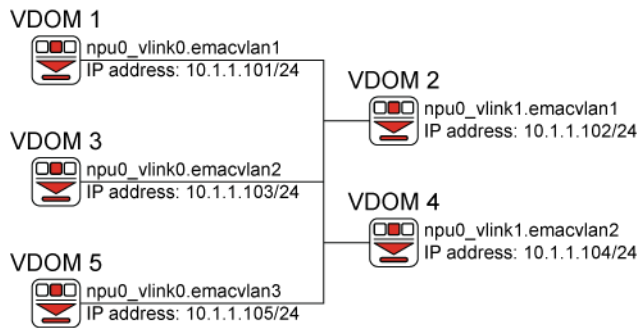
To configure enhanced MAC VLAN for this example in the CLI:

```
config system interface
  edit port1.emacvlan1
    set vdom VDOM1
    set type emac-vlan
    set interface port1
  next
  edit port 1.emacvlan2
    set vdom VDOM2
    set type emac-vlan
    set interface port1
  next
  edit port1.emacvlan3
    set vdom VDOM3
    set type emac-vlan
    set interface port1
  next
end
```

Example 2: Enhanced MAC VLAN configuration for shared VDOM links among multiple VDOMs

In this example, multiple VDOMs can connect to each other using enhanced MAC VLAN on network processing unit (NPU) virtual link (Vlink) interfaces.

FortiGate VDOM links (NPU-Vlink) are designed to be peer-to-peer connections and VLAN interfaces on NPU Vlink ports use the same MAC address. Connecting more than two VDOMs using NPU Vlinks and VLAN interfaces is not recommended as the VLAN interfaces share the same MAC address. To avoid overlapping MAC addresses on the same NPU Vlink, use EMAC VLANs instead.



To configure enhanced MAC VLAN for this example in the CLI:

```

config system interface
  edit npu0_vlink0.emacvlan1
    set vdom VDOM1
    set type emac-vlan
    set interface npu0_vlink0
  next
  edit npu0_vlink0.emacvlan2
    set vdom VDOM3
    set type emac-vlan
    set interface npu0_vlink0
  next
  edit npu0_vlink1.emacvlan1
    set vdom VDOM2
    set type emac-vlan
    set interface npu0_vlink1
  next
end

```

Example 3: Enhanced MAC VLAN configuration for unique MAC addresses for each VLAN interface on the same physical port

Some networks require a unique MAC address for each VLAN interface when the VLAN interfaces share the same physical port. In this case, the enhanced MAC VLAN interface is used the same way as normal VLAN interfaces.

To configure this, use the `set vlanid` command for the VLAN tag. The VLAN ID and interface must be a unique pair, even if they belong to different VDOMs.

To configure enhanced MAC VLAN:

```

config system interface
  edit <interface-name>
    set type emac-vlan
    set vlanid <VLAN-ID>
    set interface <physical-interface>
  end

```

```
next
end
```

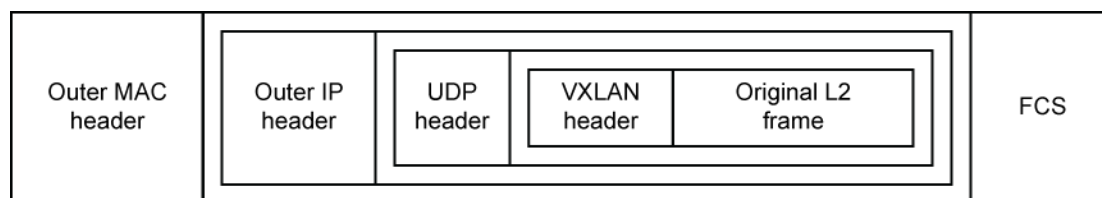


FortiGate supports a maximum of 512 EMAC VLAN interfaces per underlying interface, and a maximum of 600 MAC addresses including EMAC VLAN interfaces.

VXLAN

Virtual Extensible LAN (VXLAN) is a network virtualization technology used in large cloud computing deployments. It encapsulates layer 2 Ethernet frames within layer 3 IP packets using the UDP transport protocol on port 4789. VXLAN endpoints that terminate VXLAN tunnels can be virtual or physical switch ports, and are known as VXLAN tunnel endpoints (VTEPs).

Sample VXLAN packet



A VXLAN packet encapsulation occurs by first inserting a VXLAN header in front of the original layer 2 frame. This VXLAN header uses 3 B for the VNID that is used to identify the VXLAN segment, meaning that there are 16,777,215 different possible VNIDs. This allows for more unique LAN segments than possible VLANs. The original frame and the VXLAN header are then encapsulated into the UDP payload. The outer IP header allows it to be routed and transported over a layer 3 network, thus providing a layer 2 overlay scheme over a layer 3 network.

This equates to 50 B of overhead over the original frame: 14 B (Ethernet) + 20 B (IPv4) + 8 B (UDP) + 8 B (VXLAN headers). Since fragmenting a VXLAN packet is not recommended, it is advisable to increase the MTU size to 1550 B or above if possible, or to decrease the TCP MSS size inside a firewall policy.

For more information about VXLAN, see [RFC 7348](#).

The following topics provide information about VXLAN:

- [General VXLAN configuration and topologies on page 314](#)
- [VLAN inside VXLAN on page 318](#)
- [Virtual wire pair with VXLAN on page 320](#)
- [VXLAN over IPsec tunnel with virtual wire pair on page 322](#)
- [VXLAN over IPsec using a VXLAN tunnel endpoint on page 326](#)
- [VXLAN with MP-BGP EVPN on page 332](#)
- [EVPN support for VXLAN Anycast gateway and type-5 routing New on page 345](#)
- [VXLAN troubleshooting on page 361](#)

General VXLAN configuration and topologies

This topic describes general VXLAN configurations and commonly used topologies. In the most basic configuration, a FortiGate is configured as a VXLAN tunnel endpoint (VTEP).

To configure a FortiGate as a VTEP:

1. Configure the local interface:

```
config system vxlan
  edit <name>
    set interface <string>
    set vni <integer>
    set ip-version {ipv4-unicast | ipv6-unicast | ipv4-multicast | ipv6-multicast}
    set dstport <integer>
    set remote-ip <IP_address>
    set remote-ip6 <IP_address>
  next
end
```

interface <string>	Set the local outgoing interface for the VXLAN encapsulated traffic.
vni <integer>	Set the VXLAN network ID.
ip-version {ipv4-unicast ipv6-unicast ipv4-multicast ipv6-multicast}	Set the IP version to use for the VXLAN device and communication over VXLAN (default = ipv4-unicast).
dstport <integer>	Set the VXLAN destination port (default = 4789).
remote-ip <IP_address>	Set the IPv4 address of the remote VXLAN endpoint.
remote-ip6 <IP_address>	Set the IPv6 address of the remote VXLAN endpoint.

The VXLAN system interface is automatically created with a vxlan type.

2. Configure the VXLAN interface settings:

```
config system interface
  edit <name>
    set vdom <string>
    set type vxlan
    set ip <IP_address>
    set allowaccess {ping https ssh http telnet fgfm radius-acct probe-response fabric ftm speed-test}
  next
end
```

3. Connect the internal interface and VXLAN interface to the same L2 network.

- Connect using a software switch:

```
config system switch-interface
  edit <name>
```

```

    set vdom <string>
    set member <member_1> <member_2> ... <member_n>
    set intra-switch-policy {implicit | explicit}
  next
end

```

```

member <member_1>
      <member_2> ...
      <member_n>

```

Enter the VXLAN interface and other physical or virtual interfaces that will share the L2 network.

When adding an interface member to a software switch, it cannot have an IP address or be referenced in any other settings. For newly created VLAN interfaces, it is advised to change the role from LAN to undefined so that an address is not automatically assigned.

```

intra-switch-policy
{implicit |
explicit}

```

Allow any traffic between switch interfaces or require firewall policies to allow traffic between switch interfaces:

- **implicit:** traffic between switch members is implicitly allowed.
- **explicit:** traffic between switch members must match firewall policies (explicit firewall policies are required to allow traffic between members).

When in explicit mode, traffic can be offloaded to SOC4/SOC5/NP6/NP7 processors.

- Connect using a virtual wire pair:

```

config system virtual-wire-pair
  edit <name>
    set member <member_1> <member_2>
    set wildcard-vlan {enable | disable}
    set vlan-filter <filter>
  next
end

```

```

member <member_1>
      <member_2>

```

Enter the VXLAN interface and other physical or virtual interface that will share the L2 network.

```

wildcard-vlan {enable |
              disable}

```

Enable/disable wildcard VLAN. Disable to prevent VLAN-tagged traffic between the members of the virtual wire pair (default). Enable for VLAN tags to be allowed between the members.

```

vlan-filter <filter>

```

When wildcard-vlan is enabled, set the VLAN filter to specify which VLANs are allowed. By default, an empty vlan-filter allows all VLANs.

4. If using a virtual wire pair, configure a firewall policy that allows bi-directional traffic between the members of the virtual wire pair and inspection between them:

```

config firewall policy
  edit <id>
    set name <name>
    set srcintf <member_1> <member_2>
    set dstintf <member_1> <member_2>
    set srcaddr "all"

```